

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester IV Examination, 2023-24

Paper No. MTB-402: Mathematical Methods

Time: 3 Hours

Full Marks: 70

Ref. No.: 074887

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each unless stated.)

- (a) Using the definition of Laplace transform, find $\mathcal{L}\{\cos(at + b)\}$, $a, b \in \mathbb{R}$.
- (b) Prove that Laplace transform is linear.
- (c) Find the convolution of t and e^t .
- (d) Find the Laplace transform of $f(t) = e^t$, $0 < t < 2\pi$, $f(t + 2\pi) = f(t)$.
- (e) Write the function whose Laplace transform is $\frac{1}{s}$.
- (f) If $\mathcal{L}^{-1}[F(s)] = f(t)$, prove that $\mathcal{L}^{-1}[F(as)] = \frac{1}{a}f\left(\frac{t}{a}\right)$, $a \in \mathbb{R}$.
- (g) If $f(x)$ has Fourier coefficients a_n, b_n , show $kf(x)$ has Fourier coefficients ka_n, kb_n .
- (h) If the Fourier cosine integral of $f(x)$ is $f(x) = \int_0^\infty A(\omega) \cos(\omega x) d\omega$, prove $(-x^n)f(x) = \int_0^\infty A^{(n)}(\omega) \cos(\omega x) d\omega$. [3]
- (i) What is the Brachistochrone problem? [2]
- (j) Test for an extremum of $I[y] = \int_0^1 (xy + y'^2 - 2y^2 y') dx$, $y(0) = 4$, $y(1) = 2$. [2]
- (k) Find the Laplace transform of the Heaviside function. [2]

Question 2.

- (a) State and prove sufficient conditions for the existence of the Laplace transform. [6]
- (b) Find the Fourier sine series of the periodic function of period π :

$$f(x) = \begin{cases} \pi/4 & 0 < x < \pi/2 \\ \pi - x & \pi/2 \leq x < \pi \end{cases}$$

Hence deduce that $\frac{\pi^2}{8} = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}$. [6]

Question 3.

- (a) If $f(t)$ is continuous except for an ordinary discontinuity at $t = a$ ($a > 0$), show that

$$\mathcal{L}\left[\frac{d}{dt}f(t)\right] = s\mathcal{L}[f(t)] - f(0) - e^{-as}[f(a+0) - f(a-0)]. \quad [6]$$
- (b) Find the Fourier series of $f(x) = x^2$, $0 < x < 2$ (period 2). Hence deduce $\frac{\pi^2}{12} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$. [6]

Question 4.

- (a) If $F(s)$ is the Laplace transform of $f(t)$, find $f(t)$ where $F(s) = \frac{s}{(s^2 + a^2)(s^2 + b^2)}$. [6]
- (b) State and prove the Convolution Theorem for Laplace transform. [6]

Question 5.

- (a) Apply Laplace transform to solve: $\frac{d^2y}{dx^2} + 9y = \cos 2x$, $y(0) = 1$, $y'(\pi) = -1$. [6]
- (b) Apply Laplace transform to solve $\frac{dy}{dt} = \sinh at - 4 \int_0^t y(v) \cosh 2(t-v) dv$, $y(0) = 1$. [6]

Question 6.

- (a) Using Calculus of Variations, find the shortest distance between $(1, 0)$ and the ellipse $4x^2 + 9y^2 = 36$. [6]
- (b) Find the extremal for $I = \int_0^{\pi/2} (y''^2 - y^2 + x^2) dx$, $y(0) = 0$, $y'(0) = 1$, $y(\pi/2) = y'(\pi/2) = \frac{1}{\sqrt{2}}$. [6]

Question 7.

- (a) Find the Fourier cosine and sine integral representations of $f(x) = e^{-kx}$, $x \geq 0$, $k > 0$. [6]
- (b) Find the extremal through the origin and $(1, 1)$ of $I = \int_0^1 y^2(y'^2 - x^2) dx$ using $x^2 = u$, $y^2 = v$. [6]

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